DO NOW — RECALL

Data types & unsigned binary

Name: _____ Class: _____ Date: _____

1. Choose the most suitable data type for 27 students and for a temperature of 18.5 °C .

2. A student ID is "0042" . Which data type would you use, and why would an integer be unsuitable?

3. Convert 00101101_2 into denary. Show the place values you add.

4. A student says: “The largest unsigned 8-bit value is 256 because $2^8 = 256$.” Explain the mistake.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Integer for 27; real / floating point for 18.5. A count uses whole numbers. A temperature may contain a fractional part.

2. Model answer

String . The value is an identifier rather than a number for calculation. A string preserves the leading zeros; an integer would store the numerical value 42.

3. Model answer

45 32 + 8 + 4 + 1 = 45

4. Model answer

There are 256 different bit patterns , but the unsigned range starts at 0. Eight bits represent 0–255 . The value 256 needs 9 bits: 100000000_2 .

By the end, you can…

• choose and justify a suitable primitive data type;

• convert between denary and unsigned 8-bit binary;

• explain place values and the range of an unsigned byte.

Vocabulary — know these terms

Recall the term — Integer

Recall the term — Real / floating point

Recall the term — Character

Recall the term — String

Recall the term — Boolean

Recall the term — Bit

Recall the term — Byte

Recall the term — Unsigned binary

NEW LEARNING

Read, annotate and explain

Choose the type · read the bits

Integer whole number · Real fractional number

Character one character · String sequence of characters

Boolean True / False

8-bit unsigned weights:
128 64 32 16 8 4 2 1

Unsigned 8-bit range: 0–255. There are 256 possible patterns because zero is included.

Choose a type for the purpose

The symbols in a value do not decide its data type on their own. Think about what the value represents and what operations make sense.

Binary → denary

Select only the place values whose bits are 1.

Denary → binary

Choose place values that add to the target. Write 1 in those columns and 0 in the others.

Know the range

Eight bits give $2^8 = 256$ different patterns. The range begins at 0, so the largest unsigned value is $2^8 - 1$.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Binary → denary

Write the columns

```
128 64 32 16 8 4 2 1
 0   0  1  0 1 1 0 1
```

Use only the columns containing 1

```
32 + 8 + 4 + 1
```

Calculate the denary value

```
32 + 8 + 4 + 1 = 45
```


WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Build the justification

Guided 2 — CHECK

Choose the place values
74 = 64 + 8 + 2.
Check: 64 + 8 + 2 = 74.

Guided 3 — CHECK

Build the answer

Guided 4 — CHECK

Complete the rule

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

The largest unsigned value that can be represented using 8 bits is 255.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Choose the best type

A school stores this as a student identifier. Which data type is most suitable, and why?

Method / working:

Guided 2 — TRY: Encode 74

Write 74 as an unsigned 8-bit binary value.

Method / working:

Guided 3 — TRY: Represent an unsigned integer

Convert the denary number 188 to an unsigned 8-bit binary number. OCR June 2017 question paper .

Method / working:

Guided 4 — TRY: Why is the maximum 255?

Explain the maximum unsigned value in 8 bits and what happens when we need to store 256.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

A weather-monitoring system
For each field, state a suitable data type and justify it:
number of readings collected;
air temperature, e.g. 18.5 °C;
a one-letter category, e.g. A;
station identifier "0042" ;
whether the sensor is working.

Question 2

Encode the readings
Convert each value into unsigned 8-bit binary. Show a place-value decomposition.
CODE:
a) 37
b) 92
c) 200
Classroom adaptation of the conversion skill used in OCR June 2017 Q6(a)(i).

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Convert unsigned 8-bit binary to denary
Convert each unsigned 8-bit binary number to denary. Show which place values contribute to the total.
CODE:
a) 01101001_2
b) 11000010_2

Question 4

Choose a storage width
A system must store whole-number readings from 0 to 300 inclusive.
State the largest unsigned 8-bit value.
Explain why 8 bits are insufficient.
Determine the minimum number of bits required and justify it using the unsigned range.